



CAMSHIELD

Webcam Security & Intrusion
Detection System

Project Overview

- **Project Name:** CamShield – Webcam Security and Intrusion Detection System
- **Version:** v1.0
- **Programming Language:** Python 3.11
- **GUI Framework:** PyQt6
- **Project Type:** Cyber Security Application
- **Development Environment:** Visual Studio Code
- **Database:** JSON / SQLite
- **Status:** Completed

Problem Statement

- Unauthorized access to webcams and computer systems.
- Lack of real-time user identity verification.
- Absence of automated intruder detection mechanisms.
- No immediate notification when unauthorized access occurs.
- Insecure storage of security logs and monitoring records.
- Increased risk of privacy violations and security threats.

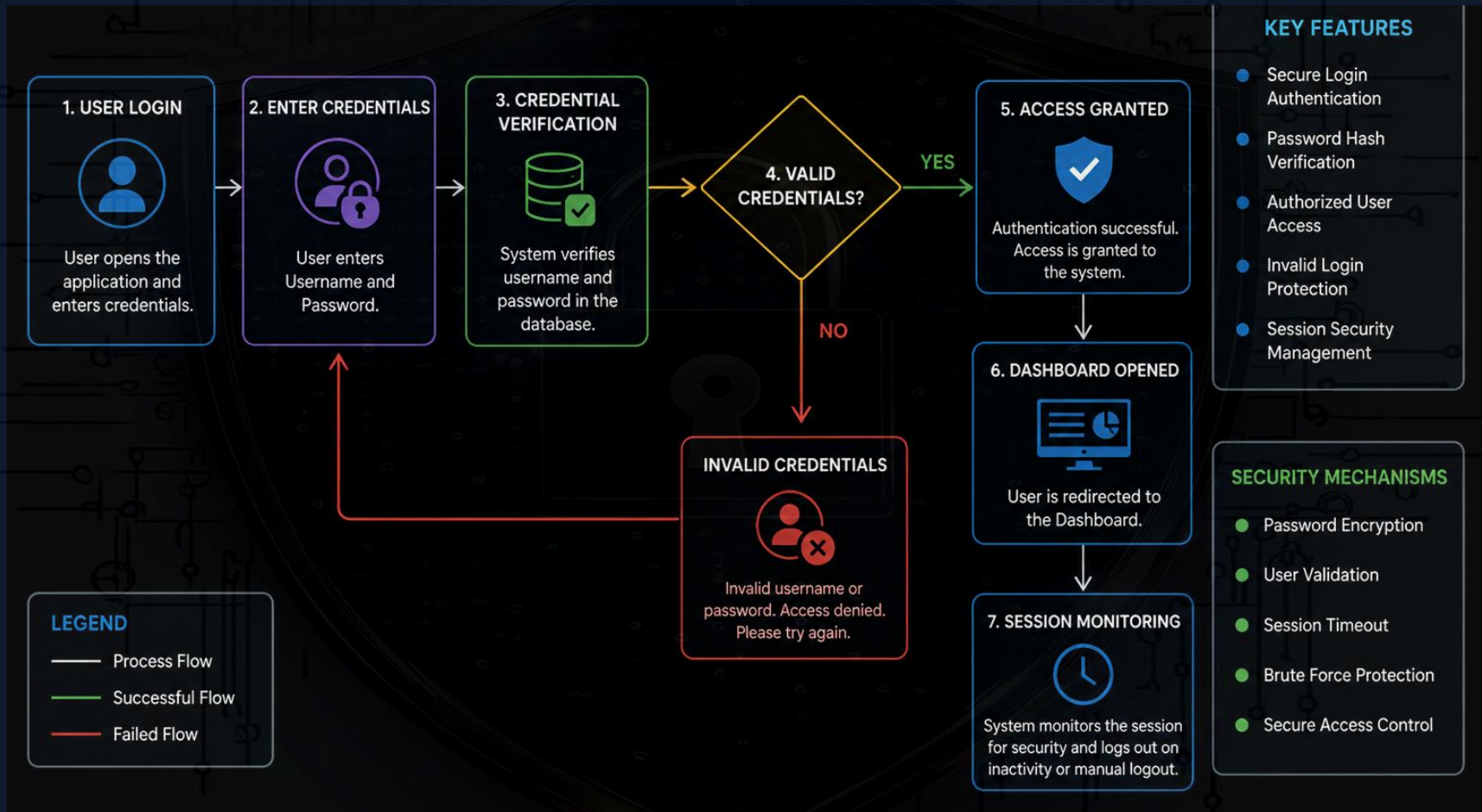
Key Features

- User Authentication
- Face Registration
- Face Recognition
- Intruder Detection
- Email Notifications
- Encrypted Logs

System Architecture

- **Presentation Layer**
 - Login Interface
 - Registration Interface
 - Dashboard
 - Face Registration Screen
 - Monitoring Panel
- **Business Logic Layer**
 - User Authentication
 - Face Recognition
 - Webcam Monitoring
 - Intruder Detection
 - Email Alert System
- **Data Layer**
 - User Database
 - Face Encodings
 - Encrypted Logs
 - Intruder Images

Authentication Flow



Face Recognition Workflow

- • Capture Webcam Frame
- • Detect Face
- • Generate Face Encoding
- • Compare with Database
- • Authorized/Unauthorized Decision

Intruder Detection

- • Unknown Face Detection
- • Capture Intruder Image
- • Store Evidence
- • Generate Logs
- • Send Email Alerts

Security Implementation

- • bcrypt Password Hashing
- • Fernet Encryption
- • Secure Key Management
- • Encrypted Log Storage
- • Access Control

Technology Stack

- • Python 3.11
- • PyQt6
- • OpenCV
- • Face Recognition
- • Cryptography
- • JSON/SQLite
- • SMTP Email Service

Testing & Results

- • Login Testing
- • Face Recognition Accuracy
- • Intruder Detection Testing
- • Email Alert Validation
- • Successful Monitoring Results

Conclusion

- • Enhanced Webcam Security
- • Real-Time Intrusion Detection
- • Reliable User Authentication
- • Secure Logging & Alerts
- • Practical Cybersecurity Solution

Future Scope

- • Mobile Notifications
- • Cloud Storage
- • AI-Based Threat Analysis
- • Multi-Factor Authentication
- • Cross-Platform Support

Development Team

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THANK YOU !